LEO Cubesat Power Management System

IEEE AESS Sustainability Hackathon 2026 | Challenge 1: Sustainable Electronics for Space Systems

Problem Statement

Cold-season eclipses force a CubeSat EPS to balance science collection, battery heating, and safe charging. A weakly protected baseline wastes heater energy or risks cold-charge exposure, degrading battery health.

Core Function

An ESP32 controller picks a mode every 60-second heartbeat from solar, SOC, temperature, and fault inputs, setting: payload rail duty cycle, battery heater pulse duty (20/35/100%), and charge enable/inhibit, gated to 0–45°C.

Baseline Case vs. Improved Case

Case	Description
Baseline — quetzal_style_heater_protected	Protected heater estimate: thermostat-latched full heater at 3°C, off at 5°C, charge allowed above 0°C.
Improved — our_adaptive_albedo	Tuned payload/heater duty schedule per 60 s heartbeat with a strict 0–45°C charge gate and lower heater draw.
Diagnostic — no_charge_temp_gate_baseline	Stress baseline with no charge-temperature gate, used only to show the value of the charge gate.

Test Conditions

Annual model period:	365.25 days	Battery capacity:	14.8 Wh
Orbit period:	94.469 min	Solar input:	2.37 W
Nominal eclipse duration:	31.5 min	Fixed bus load:	0.6637 W
Initial SOC / battery temp:	88% / 12°C	Payload active load:	0.123 W

Scenarios tested (annual, projected to 1/3/5-yr checkpoints): warm_nominal_year (31.5 min eclipse); realistic_cold_season_year (+2×45-day cold windows, 36.0 min eclipse); cold_long_eclipse_stress_year (+2×45-day windows, 41.5 min eclipse, 22°C/-20°C).

Primary Technical KPI — 5-Year Energy & Heater Savings vs. Protected Baseline

Scenario	Protected avg power	Our avg power	Energy saved	Heater saved	Battery-health proxy
warm_nominal_year	0.747 W	0.746 W	0.14%	0.00%	71.70% → 71.73%
realistic_cold_season_year	0.803 W	0.751 W	6.51%	90.48%	67.26% → 71.38%
cold_long_eclipse_stress_year	0.827 W	0.756 W	8.59%	86.73%	52.94% → 71.14%

Stress-year 5-yr totals: 36.24→33.12 kWh energy, 4.36→0.58 kWh heater, 52.9%→71.1% battery proxy.

Primary Sustainability KPI — Science Retention with Reduced Waste & Extended Battery Life

Keeps **more than 98%** of daylight science opportunity while cutting heater energy waste by up to **90.48%** and boosting the 5-year battery-health proxy by up to **18.2 points** versus the protected baseline — longer battery life, less wasted energy.

Project Limitations

Analytical model only — not real flight telemetry or a cell-qualified lifetime prediction. Needs bench thermal testing, real battery-stack data, and hardware validation before flight-equivalent claims.

Full assumptions/citations: docs/ASSUMPTIONS_PARAMETERS_AND_CITATIONS.md